

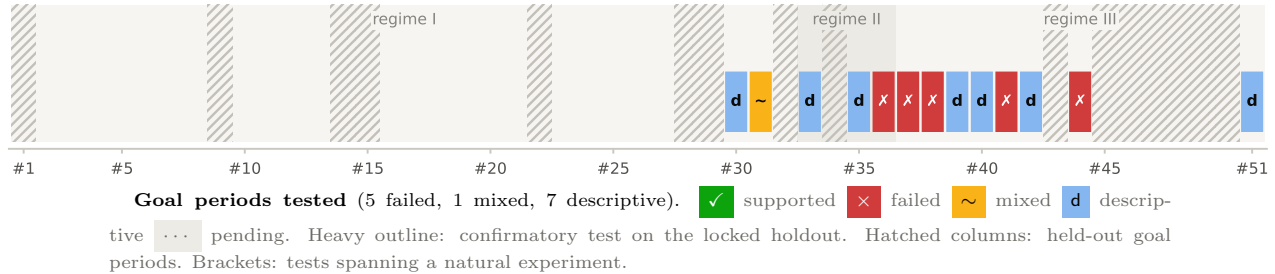
H129: Project hopping carries cycle currents: the sticky Potts walker breaks detailed balance

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 1/5 Usefulness 1/5

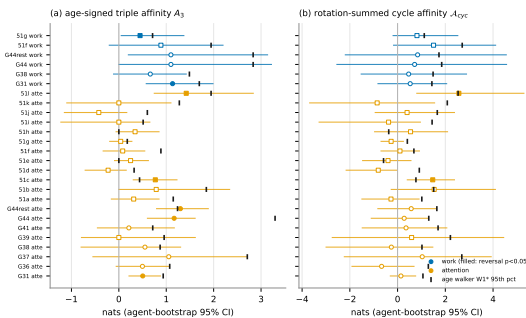
Bottom line. Agents' project hops carry no circulation beyond age drift (cycle affinity above a calibrated age walker in 0/9 shared-goal unit-channels, power 0.55–0.97); net flow to newer projects is small but consistent (30/37 positive) and dwell hazards age rather than stay geometric (32/39).

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

HH370 says that agents hopping between projects behave like a sticky Potts walker whose flows are driven around cycles ($A \rightarrow B \rightarrow C \rightarrow A$) by projects being born and dying, so detailed balance breaks with a circulation from older to newer projects, and dwell times are geometric. H129 extracts every agent’s sequence of distinct projects (work host labels and attention labels) in 13 non-holdout periods and 12 #51 units and measures triple affinities against a per-agent time-reversal null and against an age walker calibrated to each unit. **There is no circulation beyond age drift:** the rotation-summed cycle affinity beats the age walker in 0/9 shared-goal unit-channels (power 0.55–0.97 for a 2-nat cycle). HH370’s literal test passes in only 3/9. Dwell hazards age (32/39). The holdout is not run.



Observables. (a) Age-signed triple affinity A_3 and (b) rotation-summed cycle affinity $\mathcal{A}_{\text{cyc}}$ per testable unit-channel (work blue, attention orange; agent-bootstrap 95% CIs; filled markers: per-agent reversal null $p < 0.05$; circles shared-goal, squares own-role). Black ticks: 95th percentile of the age walker calibrated to the unit’s net age flux and return share. No shared-goal unit exceeds its tick in (b).

PHYSICAL MODEL

Models 10, 02 and 15. Sticky walker: per own call an agent leaves with probability r ; at a hop it picks j with $P(j) \propto e^{\alpha_j + b[\text{held}] + \lambda z_j(t)}$, z_j the standardized age rank among available projects. Any potential (fixed attractiveness, age bias λ) makes the three rotations of a triangle cancel; a cyclic force κ does not. Age drift is excess entropy production; a cycle is housekeeping.

$$A_3 = \ln \frac{N(omn) + \frac{1}{2}}{N(nmo) + \frac{1}{2}}, \quad \mathcal{A}_{\text{cyc}} = \sum_{\text{rot}} \ln \frac{N(\text{rot}) + \frac{1}{2}}{N(\overline{\text{rot}}) + \frac{1}{2}} \approx 2 \ln \frac{P_{om} P_{mn} P_{no}}{P_{on} P_{nm} P_{mo}}$$

$$m_2 = \frac{n_{\uparrow} - n_{\downarrow}}{n_{\uparrow} + n_{\downarrow}}, \quad \ln h(d) \simeq a + \gamma \ln d$$

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For an operator: agents do not rotate through projects. They drift slightly toward the newest ones (net flow 0.03–0.21 of hops) and become stickier the longer they stay (hazard falls by about 20–60% per e-fold of dwell). To move an agent off a project, act early in its stay; an agent several hours into a project rarely leaves on its own. Expect few returns-in-a-ring; plan handoffs as one-way moves to new work.

RESULTS VS PREDICTIONS

Prediction (locked)	Observed	Verdict
F: valid circulation test	$\mathcal{A}_{cyc} + W1^* 5/6$ skeletons; C_2 not identified	partly
P1 $A_3 > 0$ vs reversal (kill)	3/9 shared unit-channels	killed
P2 $\mathcal{A}_{cyc}$ beyond age 0/9 walker		failed
P3 Hodge curl	0/5; not identified (A1)	n.i.
P4 $m_2 > 0$, $p < 0.05$, $\geq 2/3$	8/34; sign + in 30/37 (post hoc)	failed
P5a geometric dwell	6/39; aging in 32/39	failed
P5b rate falls with habit	$\rho = -0.29$ (p 0.53)	weak
P6 own-role smaller	reversed (2 vs 3 units)	failed
N2 #38 drift, no cycle	m_2 0.17 (p 0.03); $\mathcal{A}_{cyc}$ inside	supported

GOAL PERIODS IN DETAIL

G30 d descriptive
G31 $\sim$ mixed
G33 d descriptive
G35 d descriptive
G36 $\times$ failed
G37 $\times$ failed
G38 $\times$ failed
G39 d descriptive
G40 d descriptive
G41 $\times$ failed

+3 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

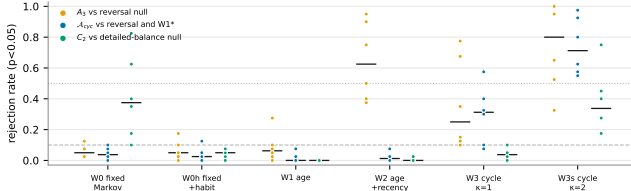
Work hops are few outside #51 (2–120 per period); most per-period verdicts rest on 30–170 triples, and weak cycles ($\kappa = 1$: power 0.07–0.57) are not excluded. Attention labels are artifact mentions, not work. Age is the first non-holdout appearance; projects started before the record look young. The m_2 sign test and #51 pooling are post hoc (A2); two own-role #51 attention units pass both nulls, about the chance rate for 14 tests.

ROUND 2

Model dwell as an aging renewal process and relate its exponent to habit and to H72's gate aging; replicate the two #51 hits across units; bound the excess EP of the age flow. Confirm C0–C3 on #45–#47.

- **H129-R1** Aging dwell as a model.
- **H129-R2** Pool cycle tests across #51's private-goal units on the work channel once more units pass the triple floor (only 2/12 now); use the 12 attention units as a replica test of the two hits (51c, 51l).
- **H129-R3** Excess bound.

Second figure. Synthetic validation on six real skeletons (40 datasets per world). A_3 against the reversal null rejects in an age walker with recency and no cycle (W2), so it measures drift. $\mathcal{A}_{cyc}$ against both nulls stays below 0.10 in every no-cycle world and detects a 2-nat cycle (W3s) in 0.55–0.97 of datasets. The detailed-balance null for the Hodge curl rejects in the reversible fixed world W0 (finite-time transients), so C_2 is not identified.



H129-project-cycle-currents · generated 2026-10-04 16:18 by in-fra/summaries/build_summaries.py · card hypotheses/H129-project-cycle-currents/README.md · ratings: H129 agent (suggested) (page 2)